

Task 7

PCB Design using eSim

LM386-Based Microphone Audio Amplifier

Name	Santhosh C
Registration No.	24VL0052
Program	B.E. Electronics Engineering (VLSI Design & Technology)
Institution	Chennai Institute of Technology
Internship	FOSSEE eSim Semester Long Internship - IIT Bombay
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EDA Tool	eSim (KiCad-based), Version 2.3 / 2.5

1. Project Overview

This project presents the complete PCB design of an **LM386-based Microphone Audio Amplifier**, developed as part of Task 7 (PCB Design using eSim) of the FOSSEE eSim Semester Long Internship. The objective of the task was to demonstrate the full PCB design workflow in the eSim (KiCad-based) environment - from schematic capture through footprint assignment, PCB layout, routing, Design Rule Checking (DRC), and 3D visualization.

The circuit chosen is a simple, low-power condenser-microphone audio amplifier built around the **LM386 low-voltage audio power amplifier IC**. It amplifies the weak electrical signal produced by a condenser microphone and drives a small loudspeaker, making it suitable for applications such as intercoms, portable audio devices, sound-level sensing, and small public-address style setups.

Key project attributes:

- Complete schematic-to-PCB flow executed entirely within eSim (KiCad) Version 2.3/2.5.
 - Two-copper-layer (F.Cu / B.Cu) PCB design with through-hole components.
 - Fully routed board with zero unrouted nets and a clean Design Rule Check (0 errors, 0 warnings).
 - 3D-rendered board view generated directly from the eSim PCB editor for physical verification.
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2. Circuit Description

The design is based on the classic **LM386 microphone audio amplifier circuit**, referenced from the CircuitDigest article *"LM386 Based Audio Amplifier Circuit"* (circuitdigest.com/electronic-circuits/lm386-audio-amplifier-circuit). The LM386 is a low-voltage audio power amplifier widely used in battery-powered devices such as radios, intercoms and toys. It offers a wide supply range of 4-12 V and an internally set voltage gain of 20, which can be raised up to 200 by connecting a capacitor (and optionally a resistor) between pins 1 and 8.

2.1 Reference Circuit

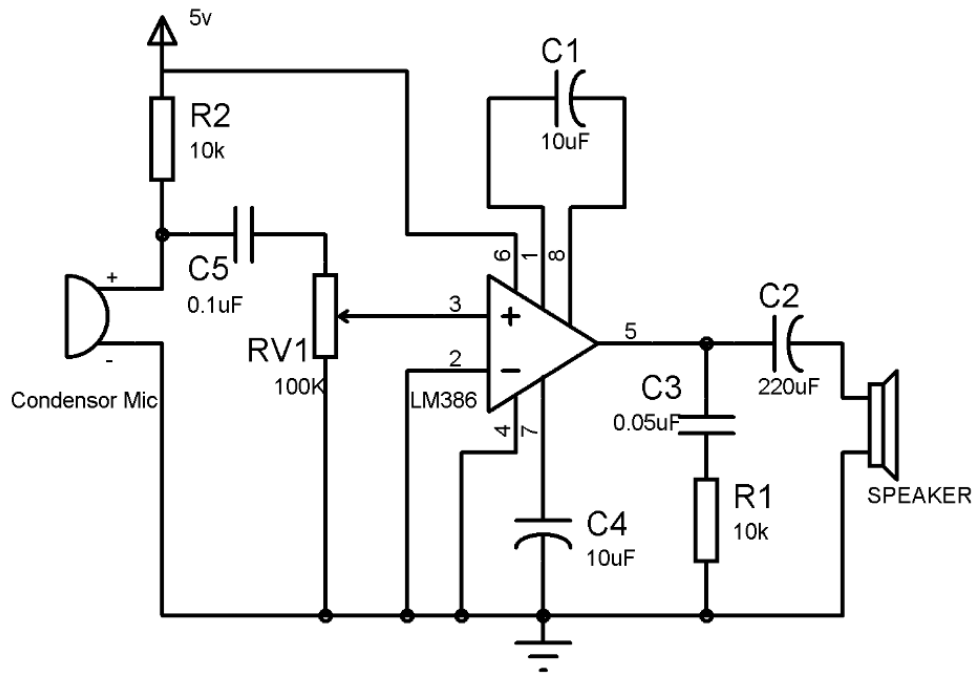


Fig. 1 - Reference LM386 microphone amplifier circuit diagram (source: CircuitDigest).

2.2 LM386 Pin Functions

The LM386 is an 8-pin DIP device. Its pin functions, used directly to build the schematic in eSim, are:

Pin(s)	Name	Function
1, 8	GAIN	Gain-set pins. Bridging with a capacitor (C1, 10uF in this design) raises the internal gain from 20 up to 200.
2 (-), 3 (+)	INPUT	Differential audio input. Pin 2 is tied to ground; Pin 3 receives the microphone signal through the volume potentiometer.
4	GND	Ground reference for the IC.
5	VOUT	Amplified audio output, fed to the speaker through a coupling capacitor and Zobel network.
6	VS	Positive supply voltage (Vcc), 5-12 V.
7	BYPASS	Internal bypass node; decoupled with a capacitor (C4) for improved supply-noise rejection.

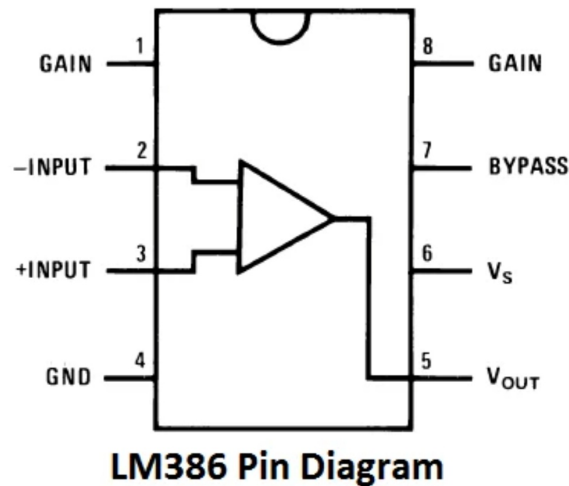


Fig. 2 - LM386 pin diagram used as design reference.

2.3 Signal Path and Component Roles

- **J1 (Mic input):** The condenser microphone connects here. R2 (10k) acts as a pull-up/bias resistor supplying power to the mic, while C5 (0.1uF) blocks the DC bias and passes only the AC audio signal into the amplifier.
- **R2 / RV1 (100k potentiometer):** Forms a volume control - the wiper feeds the adjustable audio level into the LM386's non-inverting input (pin 3).
- **C1 (10uF, pins 1-8):** Gain-set capacitor; configures the LM386 for its maximum internal gain of 200.
- **C4 (10uF) / C2 (10uF) on the ground/bypass side:** Supply and bypass decoupling for a clean, low-noise reference.
- **C3 / C5 (220uF output coupling):** Removes the DC offset from VOUT before the signal reaches the speaker.
- **R3 (10k) and C4 (0.05uF) - Zobel network:** Placed across the output to suppress high-frequency oscillation and stabilize the amplifier into the inductive speaker load.
- **J2 (Speaker) and J3 (BAT):** Terminal blocks for the loudspeaker output and the DC power supply (5-12 V) respectively.

3. Schematic Capture (eSchema)

The schematic was captured in eSim's **eSchema** editor, replicating the reference LM386 microphone amplifier circuit. All components - the LM386 IC (X1), gain/coupling/bypass capacitors (C1-C5), biasing and Zobel-network resistors (R1-R3), the volume potentiometer (R2, 100k), the microphone and speaker terminal blocks (J1, J2), and the battery terminal block (J3) - were placed and wired according to the reference design, with global VCC/GND power flags and PWR_FLAG symbols added for ERC-clean simulation readiness.

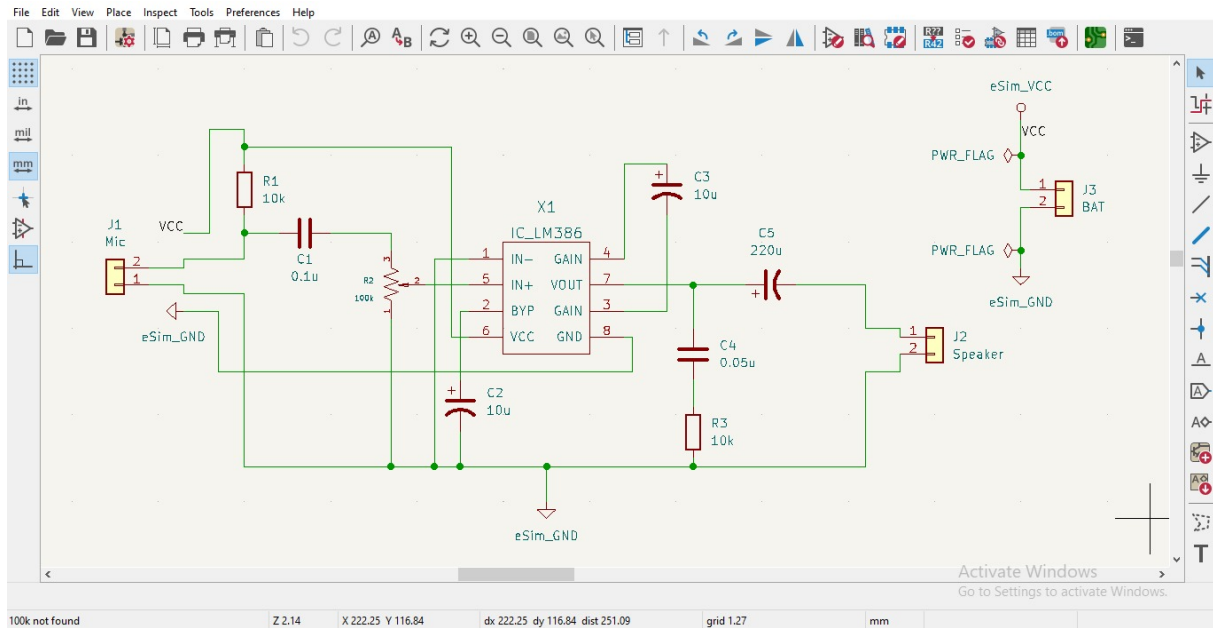


Fig. 3 - Complete schematic of the LM386 microphone audio amplifier captured in eSim eSchema.

Net labels (eSim_VCC, eSim_GND) and PWR_FLAG symbols were used to define the power nets explicitly, ensuring the schematic is electrically complete before generating the PCB via the Sch-to-PCB update flow.

4. Footprint Assignment

Each schematic symbol was mapped to a physical, through-hole footprint suited for hand assembly, using the Footprint Assignment tool. Terminal blocks (Phoenix-style, 5.08 mm pitch) were used for the microphone, speaker and battery connections; radial and disc capacitor footprints were matched to component lead spacing; and the LM386 was assigned a standard 8-pin DIP-8 package.

Symbol : Footprint Assignments		
1	C1 -	0.1u : Capacitor_THT:C_Disc_D5.0mm_W2.5mm_P5.00mm
2	C2 -	10u : Capacitor_THT:CP_Radial_D5.0mm_P2.00mm
3	C3 -	10u : Capacitor_THT:CP_Radial_D5.0mm_P2.00mm
4	C4 -	0.05u : Capacitor_THT:C_Disc_D5.0mm_W2.5mm_P5.00mm
5	C5 -	220u : Capacitor_THT:CP_Radial_D5.0mm_P2.00mm
6	J1 -	Mic : TerminalBlock_Phoenix:TerminalBlock_Phoenix_MPT-0,5-2-2.54_1x02_P2.54mm_Horizontal
7	J2 -	Speaker : TerminalBlock_Phoenix:TerminalBlock_Phoenix_MPT-0,5-2-2.54_1x02_P2.54mm_Horizontal
8	J3 -	BAT : TerminalBlock_Phoenix:TerminalBlock_Phoenix_MPT-0,5-2-2.54_1x02_P2.54mm_Horizontal
9	R1 -	10k : Resistor_THT:R_Axial_DIN0207_L6.3mm_D2.5mm_P10.16mm_Horizontal
10	R2 -	100k : Potentiometer_THT:Potentiometer_Bourns_3296W_Vertical
11	R3 -	10k : Resistor_THT:R_Axial_DIN0207_L6.3mm_D2.5mm_P10.16mm_Horizontal
12	X1 -	IC_LM386 : Package_DIP:DIP-8_W7.62mm

Fig. 4 - Symbol-to-footprint assignment table generated in eSim.

Ref	Value / Part	Assigned Footprint
C1	0.1uF	Capacitor_THT: C_Disc_D5.0mm_W2.5mm_P5.00mm
C2, C5	10uF / 220uF	Capacitor_THT: CP_Radial_D5.0mm_P2.00mm
C3	10uF	Capacitor_THT: CP_Radial_D5.0mm_P2.00mm
C4	0.05uF	Capacitor_THT: C_Disc_D5.0mm_W2.5mm_P5.00mm
J1, J2, J3	Mic / Speaker / BAT	TerminalBlock_Phoenix_MPT-0,5-2-2.54_1x02_P2.54mm_Horizontal
R1, R3	10k	Resistor_THT: R_Axial_DIN0207_L6.3mm_D2.5mm_P10.16mm_Horizontal
R2	100k (Potentiometer)	Potentiometer_Bourns_3296W_Vertical
X1	IC LM386	Package_DIP: DIP-8_W7.62mm

5. PCB Layout and Routing Methodology

After footprint assignment, the netlist was transferred to the PCB editor using the Sch-to-PCB Update function. Footprints were arranged with the connector-based components (Mic, Speaker, BAT, Potentiometer) placed toward the board edges for easy external wiring, and the LM386 IC placed centrally to minimize average track length to its surrounding passive components.

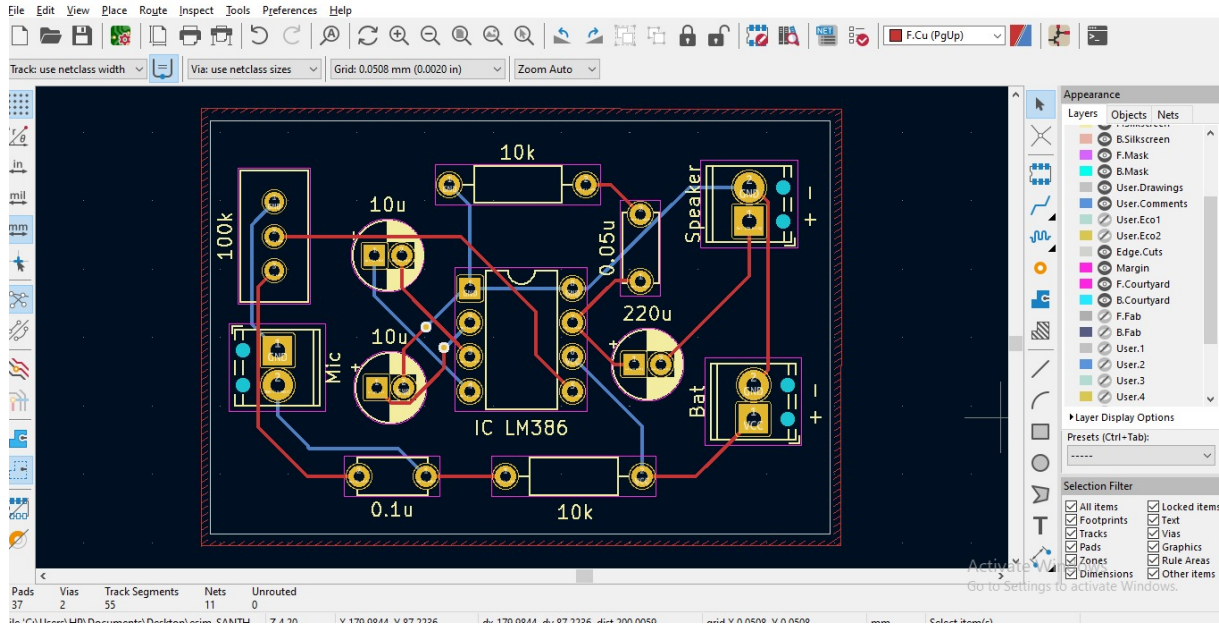


Fig. 5 - Routed PCB layout (top copper view) showing component placement and track routing.

5.1 Routing Methodology

- **Layer strategy:** A two-copper-layer board (F.Cu / B.Cu) was used. Signal tracks were primarily routed on the front copper layer, with the back copper layer used to resolve crossovers.
- **Manual interactive routing** was used for all 11 nets, following the schematic connectivity and keeping gain-set, feedback and Zobel-network traces short to minimize parasitic coupling and noise pickup around the LM386.
- **Power and ground nets** (VCC/BAT and GND) were routed with wider, direct paths between the battery terminal block, the IC supply pins, and the decoupling/bypass capacitors.
- **Track widths and clearances** followed the default eSim/KiCad netclass rules, keeping the design compatible with standard PCB fabrication tolerances.
- **Component placement** grouped input-side components (Mic, C5, R2/potentiometer) away from output-side components (speaker, C2/C3, R1) to reduce the risk of output-to-input feedback coupling.

6. Design Rule Check (DRC) Results

A full Design Rule Check was run in the eSim PCB editor after routing was completed, covering clearances, track widths, via annular rings/diameters, board outline, silkscreen overlap, and length-constrained connections.

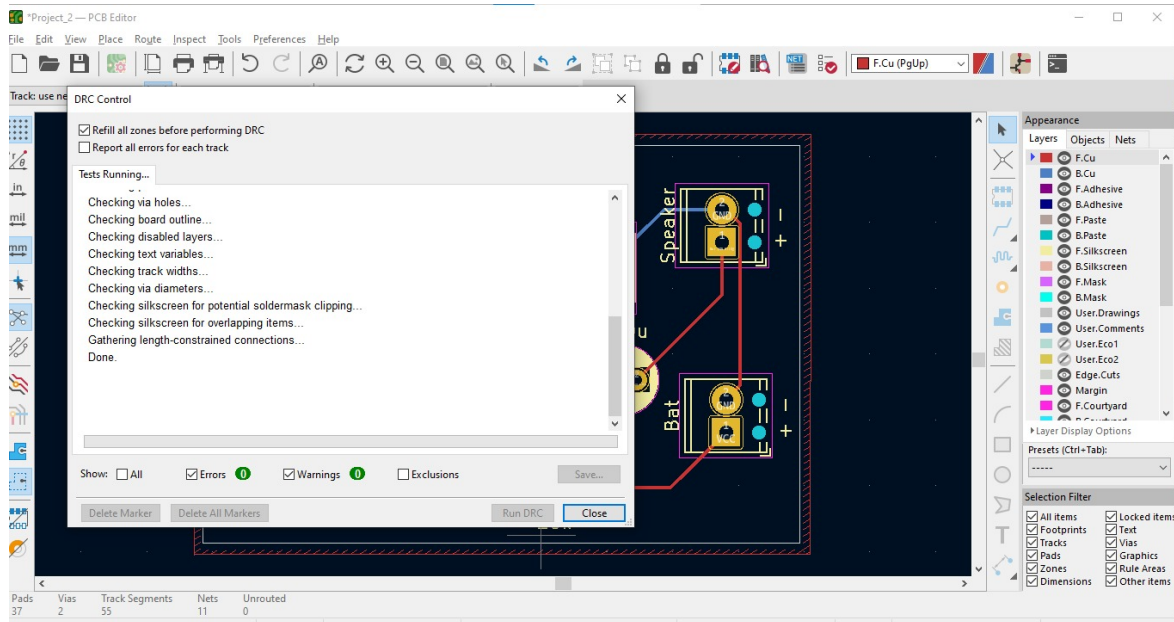


Fig. 6 - DRC Control dialog showing a completed run with 0 errors and 0 warnings.

Parameter	Result
Errors	0
Warnings	0
Pads	37
Vias	2
Track segments	55
Nets	11
Unrouted nets	0

The DRC run completed with **zero critical errors and zero warnings**, with all 11 nets fully routed (0 unrouted), confirming the board meets basic manufacturability requirements for clearance, track width, and via geometry.

7. 3D Visualization

The completed board was verified visually using eSim's built-in 3D Viewer, which renders the populated PCB from the assigned footprints and layer stack-up. This step confirms correct component orientation, footprint scale, and overall board layout before physical fabrication.

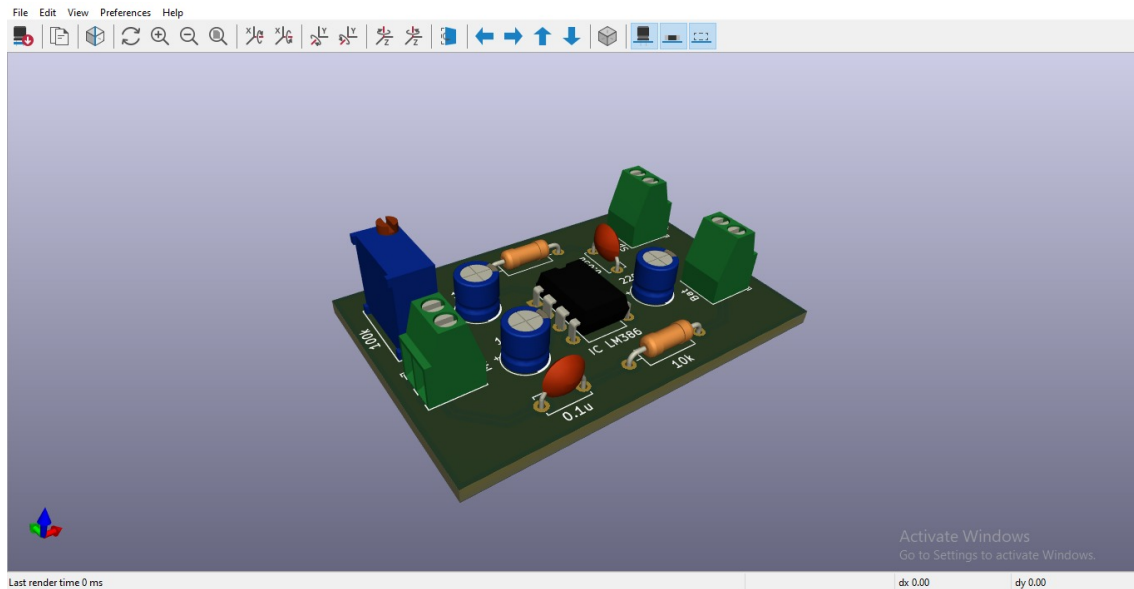


Fig. 7 - 3D isometric render of the populated PCB, showing the LM386 IC, capacitors, resistors, potentiometer and terminal blocks.

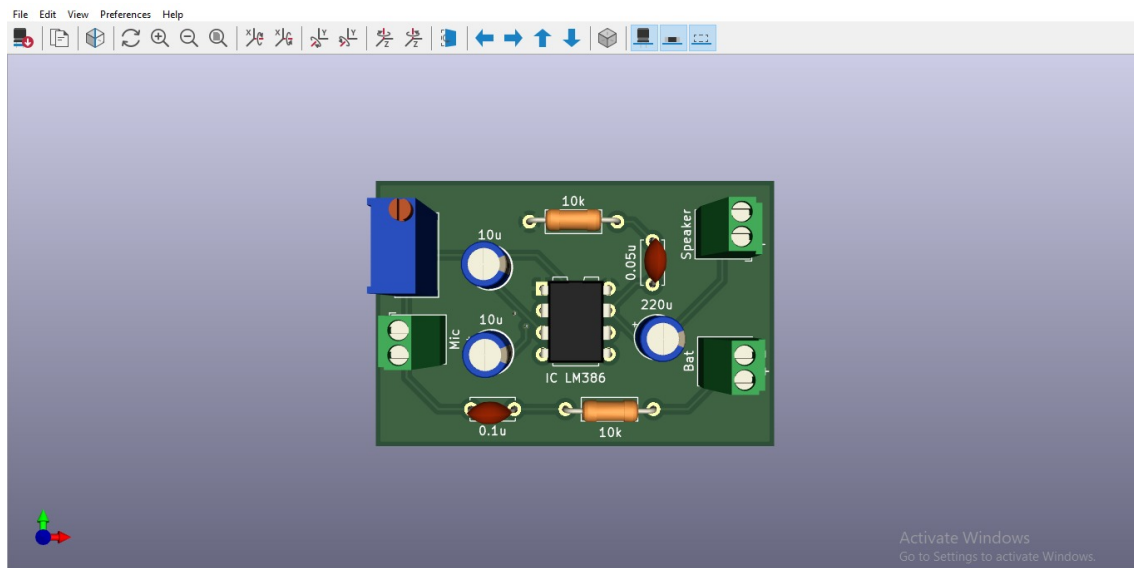


Fig. 8 - 3D top-down render of the PCB confirming component placement and silkscreen labeling.

8. Design Considerations and Conclusions

8.1 Design Considerations

- **Noise and stability:** The Zobel network (R1, C3) and bypass capacitor (C4/pin 7) were retained from the reference design to keep the amplifier stable into the inductive speaker load and to suppress high-frequency oscillation, a known sensitivity of the LM386 at high gain settings.
- **Signal integrity:** Input-side (microphone/gain) and output-side (speaker) traces were kept physically separated on the board to minimize the risk of output-to-input feedback and audio hum.
- **Serviceability:** Terminal-block connectors were chosen for the microphone, speaker, and battery inputs rather than direct solder pads, allowing easy field wiring and replacement.
- **Manufacturability:** Standard through-hole footprints and a two-copper-layer stack-up were used throughout, keeping the design compatible with low-cost PCB fabrication and hand assembly.
- **Verification:** A zero-error / zero-warning DRC result and a fully routed netlist (0 unrouted connections) confirm the layout is electrically and geometrically sound.

8.2 Conclusion

This task successfully demonstrates the complete PCB design workflow in eSim - from schematic capture in eSchema, through footprint assignment, PCB layout generation via Sch-to-PCB update, interactive routing, Design Rule Checking, and 3D visualization. The resulting LM386 microphone audio amplifier PCB is fully routed across 11 nets with zero DRC errors or warnings, validating both the correctness of the schematic and the manufacturability of the final board layout. The project confirms proficiency with the eSim (KiCad-based) PCB design environment and good PCB design practices such as signal separation, stabilization networks, and connector-based serviceability.

Reference: CircuitDigest, "LM386 Based Audio Amplifier Circuit" -
circuitdigest.com/electronic-circuits/lm386-audio-amplifier-circuit